

Is there an intraday momentum effect in commodity futures and options:  
Evidence from the Chinese Market

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ABSTRACT

Based on high-frequency data of China's commodity market from 2017 to 2022, this article examines the intraday momentum effect. The results indicate that China's commodity futures and options have significant intraday reversal effects, and the overnight opening factor and opening to last half hour factor are more significant. These effects are driven, in part, by liquidity factors. This trend aligns with market makers' behavior, passively accepting orders during low liquidity and actively closing positions amid high liquidity. Furthermore, our examination of cross-predictive ability shows strong futures-to-options predictability, while the reverse is weaker. We posit options traders' Vega hedging as a key factor in this phenomenon, our study finds futures volatility changes can predict options' return.

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#### ABSTRACT

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## 1. Introduction

Gao et al. (2018) highlight the significance of intraday momentum by demonstrating the predictive power of the overnight opening factor on final half-hour returns in US markets. Building on this, Baltussen et al. (2021) expand the scope, discovering that the rest-of-the-day factor can also effectively forecast returns in the last 30 minutes of trading. Our study aligns with the methodologies of these papers, conducting a comprehensive analysis of both factors and additional intraday momentum elements to ascertain which factor exhibits superior predictive ability for final half-hour trading returns in Chinese commodity markets. China's commodity exchanges are rapidly evolving into vital global hubs for commodity trading. According to data from the Futures Industry Association<sup>1</sup>, global futures markets traded a volume of 55.782 billion contracts in the first half of 2023. China's futures market accounted for 7.08% of this global volume. Among mainland Chinese futures exchanges, Zhengzhou Commodity Exchange (ZCE), Dalian Commodity Exchange (DCE), Shanghai Futures Exchange (SHFE), ranked 6th, 8<sup>th</sup> and 9th, respectively, in terms of global trading volumes for futures and options among all exchanges worldwide. Moreover, acknowledging the intimate link between commodity futures and options, this paper extends its scope to explore the commodity options market, analyzing the intricate interplay between these two domains.

Based on the research method of Baltussen et al. (2021), we examine the intraday momentum effect in the futures and options markets from 2017 to 2022. Both China's commodity futures market and options market show significant intraday reversal effects. This result is different with Baltussen et al. (2021), we explore the Chinese market-maker system to elucidate this phenomenon. It contends that market makers in China's commodity market adopt a strategy of closing intraday positions during high liquidity to manage risks and amplify market activity during low liquidity, displaying a unilateral trading behavior. Consequently, it categorizes futures trading into high and low liquidity groups. Through regression analysis, it establishes that China's commodity futures and options exhibit pronounced intraday reversals amid high liquidity and weaker trend characteristics during low liquidity periods, which is consistent with Chinese commodity market makers' behavior. It is worth mentioning that there are also market makers in the US market, but futures show an

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<sup>1</sup> <https://www.fia.org/fia/etd-tracker>

intraday momentum effect. This distinction may arise from the difference between U.S. market makers who operate across various major asset classes and their Chinese counterparts, who are constrained to facilitating trading solely within the commodities asset class. This disparity results in distinct risk preferences among Chinese and American market makers when dealing with a singular asset class.

In addition, we explore the cross-forecasting capabilities of China's futures and options markets. Futures have a strong cross-predictive ability to options, while the cross-predictive ability of options to futures is weak. Under the condition of controlling futures factors, call options can show significant futures price prediction ability. Regarding the source of the cross-prediction ability between futures and options, we believe that changes in the volatility of commodity futures will lead to price changes in commodity options. This paper constructs a research method for the intraday momentum effect of volatility and uses this method to explore whether changes in commodity futures volatility can predict price changes in commodity options. The results show that such predictability exists in the Chinese commodity market and is dominated by the intraday momentum effect.

Momentum effect is one of the hot phenomena studied in academia. Jegadeesh and Titman (1993) find that the "buy winners sell losers" trading strategy can achieve significant excess returns in the stock market, formally introducing the "momentum effect". Jegadeesh and Titman (2001) further strengthen evidence of the existence of the momentum effect. Rouwenhorst (1998) use the same trading strategy to demonstrate the existence of similar momentum effects in European countries. Subsequently, many studies have demonstrated the existence of momentum effects in markets such as the UK and Australia (Tonks and Hon, 2003; Drew et al., 2007, etc.). In addition to stock markets, Subrahmanyam (2018) reviews the literature on equity market momentum. Asness et al. (2013) provide evidence for eight different asset and regional markets. Chui et al. (2023) find evidence of significant price momentum (intermediate and long-term) for equities listed on the BSE. Yang et al. (2018) document strong return predictability for continuing overreaction in Taiwan. Many researchers have also focused on the momentum effect in the Chinese stock market, and their findings consistently indicate a significant momentum effect in the short term (Kang et al. 2002; Naughton et al. 2008), but an eventual reversal in the long term (Yan and Jiang, 2018; Huang et al. 2023). Researchers have also studied factors that reflect

momentum. Li et al. (2022) propose an indicator of CB to reflect investors' short-term consistent belief in price movements. Yin and Wei (2020) investigate the predictive power of aggregate profit instability for time variations in momentum returns. Momentum effect also exists in commodity market, Fan and Qiao (2023) find that commodity momentum strategies deployed in the U.S. and Chinese markets generate positive average returns. Kwon et al. (2020) suggest that the weekly momentum exist in the U.S. commodity futures market. Ming et al. (2023) find that time series momentum (TSM), a well-documented market anomaly, exists in China's financial markets. Some literature also finds similar conclusions (Zhang et al. 2020; Zhu et al. 2023).

In recent years, there has been a growing popularity surrounding intraday momentum research as a subset within the broader field of momentum research. It can be traced back to Heston et al. (2010), who explore the intraday momentum effect by dividing the daily trading time into 13 phases from cross-sectional data of the stock market. It is also found that the persistence effect was most pronounced in the first half hour and the last half hour. Li et al. (2015) examine the momentum effect model for the 1-minute to 2-day observation interval period. From time series data of the market, Gao et al. (2018) conduct the first research analysis of an intraday momentum strategy based on the S&P 500 ETF, verifying that the returns in the first half hour after the market open and the penultimate half hour before the market close can be used to predict the last half hour before the close and elaborating on the theoretical and economic aspects. Based on Gao et al. (2018), several scholars have conduct research on the intraday momentum effect in domestic and international stock and futures markets. Ho et al. (2023) analyze whether the information contained in past intraday and overnight returns affects future stock returns in different ways in Taiwan. Limkriangkrai et al. (2023) examine the market intraday momentum, where the first half-hour return predicts the last half-hour return, in exchange-traded funds (ETFs) from the selected Asia-Pacific (APAC) markets including China, Hong Kong SAR, Japan, Singapore and South Korea.

Focus on Chinese markets, Ling et al. (2017) verify the intraday momentum of Chinese CSI 300 stock index futures and CSI 500 stock index futures. Li et al. (2020) improve the time segmentation of the model by dividing the model into 15, 30 and 60 minutes to explore the intraday momentum effect of CSI 300 stock index futures. Zhang (2019) validate the intraday momentum effect on the Chinese stock market and find that both the first half hour and the seventh half hour have significant predictive effects on the last half hour returns. Specifically, the Chinese stock

market is more predictable in terms of statistical results and economic significance intraday than is the U.S. stock market. In addition, there is an opposite finding that the seventh half-hour has a stronger predictive effect than the first half-hour. Wang and Ma (2021) verify the predictive effects of the first half hour and the seventh half hour on the last half hour returns for the SSE Composite Index and a tradable index, the SSE 50 ETF, and note that the first half hour and the seventh half hour contain different predictive information. Gao et al. (2019) find that there exists a strong intraday momentum effect at Chinese firm level. The intraday predictability stems mainly from the overnight component rather than the opening half-hour component, which is consistent with the microstructure features of the Chinese market. Li et al. (2020) investigate the intraday momentum in the Chinese stock index futures market.

There are some literatures focus on Chinese future market. Jin et al. (2020) investigate intraday time-series momentum across four Chinese commodity futures contracts: copper, steel, soybean, and soybean meal. Their results indicate that the first half-hour return positively predicts the last half-hour return across all four futures. Zhang et al. (2020) construct an open-interest-weighted index with the high-frequency data of all commodity futures traded and then examine the predictability of the last half-hour return with both in-sample and out-of-sample tests. Wen et al. (2021) find a strong intraday reversal effect rather than a conventional intraday momentum effect on China's crude oil futures market. The previous night's return can significantly predict the day's return both in- and out-of-sample. However, existing research on commodities has ignored the rapidly growing market of commodity options and has not analyzed the intraday interrelationship between commodity futures and options. This study fills this gap and addresses certain shortcomings in this field.

This paper offers significant contributions to the literature by: exploring intraday momentum in Chinese commodity futures and options; introducing a liquidity-based regression addressing differences between Chinese and international markets, possibly stemming from the market-maker system; bridging gaps in discussing intraday cross-momentum effects between commodity futures and options; proposing a series of volatility momentum factors akin to the yield momentum factor to describe volatility-based momentum; and identifying the Vega Hedge behavior of options traders as a source of cross-momentum between commodity futures and options.

The remainder of the paper is organized as follows. In Section 2, we introduce the data used

and the research methods; Section 3 is an empirical analysis of the intraday momentum effect and is divided into 5 subsections, intraday momentum effect of futures, intraday momentum effect of options, liquidity and reversal effect in China's commodity market, cross-prediction relationship between futures and options, and intraday momentum characteristics of futures volatility and option prices. The last section concludes this paper.

## 2. Data and Intraday Factors

We use intraday 1-minute-level trading prices to calculate half-hourly returns. Only active commodity assets are included, and the included underlying assets are shown in Appendix A. In addition, we exclude daily data with fewer than 200 valid trading points, and the data sample period is from January 2017 to January 2022.

Compared to the near 24-hour trading hours in the U.S. futures market, the Chinese futures market is divided into two trading periods, i.e., day trading for all futures and night trading for some varieties, which are more closely related to the trading hours of the underlying stock index futures. We refer to Baltussen (2021) and establish ON, FH, M, SLH and LH times, i.e., the 24 hours from the previous day's close to today's close are divided into five hourly segments using common market trading hours. Specifically, the period between the previous day's close and today's open is defined as overnight (ON); thirty minutes after the open is defined as the first half-hour (FH); from thirty minutes after the open to one hour before the close is defined as mid-day (M); the penultimate half-hour before close is set as the second to last half-hour (SLH); and the last half-hour before the close is set as the last half-hour (LH). The five periods are further combined according to trading conditions. The combination of the previous day's close to half an hour after today's opening is called ONFH. The combination of the previous day's close to half hour before today's close is defined as the rest of day (ROD). Trading hours in the Chinese futures market are exchange- and species-related. The daily trading hours of the DCE, SHFE, ZCE and INE are from 9:00 to 15:00, including a small break from 10:15 to 10:30 and a lunch break from 11:30 to 13:30. The trading hours of the CFFEX are from 9:15 to 15:00, including a lunch break from 11:30 to 13:30. Some varieties of DCE, SHFE, ZCE and INE have night trading. Chinese options trading hours are mainly from 9:30 to 15:00, including a lunch break from 11:30 to 13:30. Therefore, the day trading hours of the futures market are selected as the trading period of the sample and fine-tuned according to

different futures markets and varieties.

To verify the predictability of intraday returns within the futures and options markets, this paper defines the returns of ONFH, M, SLH, and ROD as follows, where o represents the opening and c represents the closing:

$$r_{ONFH,t} = \frac{P_{o+30,t}}{P_{c,t-1}} - 1$$

where  $P_{c,t-1}$  represents the price at which the underlying asset was purchased at the close of the previous day,  $P_{o+30,t}$  represents the price at which the underlying asset was sold thirty minutes after the opening of the current day, and  $r_{ONFH,t}$  represents the gain from buying and holding from the close of the previous day until thirty minutes after the opening of the current day.

$$r_{ROD,t} = \frac{P_{c-30,t}}{P_{c,t-1}} - 1$$

where  $P_{c,t-1}$  represents the price at which the underlying asset was purchased at the close of the previous day,  $P_{c-30,t}$  represents the price at which the underlying asset was sold thirty minutes before the close of the day, and  $r_{ROD,t}$  represents the gain from purchasing at the close of the previous day and holding until thirty minutes before the close of the day.

$$r_{M,t} = \frac{P_{c-60,t}}{P_{o+30,t}} - 1$$

where  $P_{o+30,t}$  represents the price at which the underlying asset was purchased thirty minutes after the opening of the day,  $P_{c-60,t}$  represents the price at which the underlying asset was sold sixty minutes before the closing of the day, and  $r_{M,t}$  represents the gain from purchasing the underlying asset thirty minutes after the opening of the day and holding it until sixty minutes before the closing of the day.

$$r_{SLH,t} = \frac{P_{c-30,t}}{P_{c-60,t}} - 1$$

where  $P_{c-60,t}$  represents the price at which the underlying asset was purchased sixty minutes before closing,  $P_{c-30,t}$  represents the price at which the underlying asset was sold thirty minutes before closing, and  $r_{SLH,t}$  represents the gain from purchasing the underlying asset sixty minutes before closing and holding it for half an hour before selling it thirty minutes before closing.

### 3. Intraday momentum

To examine the intraday momentum effect in the Chinese market, we initially adopt the



approach outlined by Baltussen et al. (2021). Using a benchmark regression model, we verify the predictive impact of various predictors on the final half-hour returns ( $r_{LH,t}$ ) in a sequential manner.

First,  $r_{ONFH,t}$  is added as an independent variable to the benchmark regression to verify the predictability of overnight returns on the half-hourly returns before the close:

$$r_{LH,t} = \alpha + \beta_{ONFH} \times r_{ONFH,t} + \varepsilon_t$$

Second, the midday return ( $r_{M,t}$ ) and the penultimate half-hourly return before the close ( $r_{SLH,t}$ ) are added to the benchmark regression model (1), and the superimposed predictive effects of multiple predictors are examined in combination:

$$r_{LH,t} = \alpha + \beta_{ONFH} \times r_{ONFH,t} + \beta_M \times r_{M,t} + \beta_{SLH} \times r_{SLH,t} + \varepsilon_t$$

Third, the predictors for the overnight, midday, and penultimate half-hour periods are combined, and the rest of day ( $r_{ROD,t}$ ) is included in the benchmark regression model as the only predictor:

$$r_{LH,t} = \alpha + \beta_{ROD} \times r_{ROD,t} + \varepsilon_t$$

Fourth, the horse race between overnight return and rest of day return:

$$r_{LH,t} = \alpha + \beta_{ONFH} \times r_{ONFH,t} + \beta_{ROD} \times r_{ROD,t} + \varepsilon_t$$

To test the predictive effect of the model on out-of-sample historical data, reference is made to Gao et al. (2018) and Baltussen et al. (2021), who compared the estimates of the benchmark regression model with the mean of the penultimate half hour before the close and defined  $R_{OOS}^2$  as follows:

$$R_{OOS}^2 = 1 - \frac{\sum_{t=1}^T \frac{\sum_{f \in F_t} (r_{LH,f,t} - \hat{r}_{LH,f,t})^2}{n(F_t)}}{\sum_{t=1}^T \frac{\sum_{f \in F_t} (r_{LH,f,t} - \bar{r}_{LH,f,t})^2}{n(F_t)}}$$

### 3.1 Intraday Momentum Effect

#### 3.1.1 Commodity Future

Table 1. reports the regression results of the intraday momentum effect in the Chinese futures market, where there is a significant intraday reversal effect. This effect is not only established in pooled regression tests but also has strong predictive power in out-sample tests. The coefficient of  $r_{ONFH}$  in model (1) is negative (-0.007) and significant at the 1% level, indicating that  $r_{ONFH}$  has a

certain reverse prediction effect on  $r_{LH}$  in the Chinese futures market. This is consistent with existing research, such as Zhang et al. (2020). In model (2), the coefficients of  $r_{ONFH}$ ,  $r_M$  and  $r_{SLH}$  are all negative and significant at the 1% level. The absolute value of the coefficient of  $r_{ONFH}$  increases, the t value increases significantly, and the  $R^2$  of the equation increases from 0.048% to 0.356%, indicating that the returns of the three periods can be used to predict  $r_{LH}$ . The returns of any period since the close of the previous day contain effective prediction information for the last half-hour earnings. The  $r_{ROD}$  coefficient in model (3) is negative (-0.014) and is significant at the 1% level, indicating that  $r_{ROD}$  also has predictive ability. In addition, the  $R_{OOS}^2$  values of model (1) to model (3) are all greater than 0, namely, 0.245, 0.0008, and 0.425, respectively, indicating that  $r_{ONFH}$ ,  $r_M$ ,  $r_{SLH}$  and  $r_{ROD}$  all have predictive capabilities that exceed the historical data average. By comparing models (1)-(3), we find that  $r_{ROD}$  has the best out-of-sample prediction effect. Model (4) takes  $r_{ONFH}$  and  $r_{ROD}$  as independent variables and gives the horse racing results of  $r_{ONFH}$  and  $r_{ROD}$ . The regression results show that  $r_{ROD}$  has the greatest predictive ability.

Table 1

Regression results of the intraday momentum effect in the futures market

|             | 1                      | 2                      | 3                      | 4                       |
|-------------|------------------------|------------------------|------------------------|-------------------------|
| $r_{ONFH}$  | -0.007***<br>(-3.5350) | -0.009***<br>(-4.1600) |                        | 0.0025<br>(1.6015)      |
| $r_M$       |                        | -0.018***<br>(-8.6500) |                        |                         |
| $r_{SLH}$   |                        | -0.011*<br>(-1.6650)   |                        |                         |
| $r_{ROD}$   |                        |                        | -0.014***<br>(-9.2190) | -0.0076***<br>(-6.6304) |
| $R^2$       | 0.2450                 | 0.0008                 | 0.4250                 | 0.0701                  |
| $R_{adj}^2$ | 0.0480                 | 0.3560                 | 0.3240                 | 0.0675                  |
| $R_{OOS}^2$ | 0.0440                 | 0.3440                 | 0.3200                 | -0.0059                 |

This table reports the regression results of the return on last half hour of the futures market,  $r_{LH}$ . The meaning of each dependent variable in the table is as follows:  $r_{ONFH}$  represents the return rate of buying from the closing price of the previous day and holding until thirty minutes before the opening of the day;  $r_{ROD}$  represents the return rate of buying from the closing price of the previous day and holding until thirty minutes before the closing of the day.  $r_M$  represents the return rate from buying and holding 30 minutes after the opening of the day to 60 minutes before closing;  $r_{SLH}$  represents the return rate from buying and holding 60 minutes before closing to 30 minutes before closing;  $R^2$  represents the R-squared statistic of the linear regression; and  $R_{OOS}^2$  represents the R-squared statistic of the out-of-sample test. The sample interval is from January 2017 to January 2022. The significance levels of 1%, 5%, and 10% are marked by \*\*\*, \*\*, and \*, respectively.

### 3.1.2 Commodity Options

Commodity futures are the underlying of commodity options, so do commodity options also have significant intraday reversal effects? This section answers this question. When futures prices change, the price changes of call options and put options are not consistent; therefore, regarding the analysis of the reversal effect of options, we divide options into two groups, call options and put options, and performs regression analysis on each group. Table 2. and Table 3. report the regression results of the intraday momentum effect of Chinese commodity options. Table 2. reports the regression results of commodity call options, and Table 3. reports the regression results of commodity put options. Similar to commodity futures, commodity options have significant intraday reversal effects.

According to Table 2., the factors  $r_{ONFH}$ ,  $r_M$  and  $r_{ROD}$  all show significant reversal effects within the sample; the factors  $r_{ONFH}$  and  $r_{ROD}$  are significant at the 1% level, while  $r_M$  is significant at the 5% level. In addition, the regression results of model (4) show that  $r_{ROD}$  has a stronger predictive effect than factor  $r_{ONFH}$ . However, the performance of these factors outside the sample is very poor, and the predictive power is weaker than the mean prediction.

Table 2

Commodity call options intraday momentum effect

| CALL Option  | 1                       | 2                       | 3                       | 4                       |
|--------------|-------------------------|-------------------------|-------------------------|-------------------------|
| $r_{ONFH}$   | -0.0184***<br>(-2.8042) | -0.0192***<br>(-2.9073) |                         | -0.0100<br>(-1.3900)    |
| $r_M$        |                         | -0.0071*<br>(-1.8350)   |                         |                         |
| $r_{SLH}$    |                         | 0.0135<br>(1.0469)      |                         |                         |
| $r_{ROD}$    |                         |                         | -0.0125***<br>(-3.7152) | -0.0103***<br>(-2.7926) |
| $R^2$        | 0.0390                  | 0.0611                  | 0.0684                  | 0.0777                  |
| $R^2_{adj.}$ | 0.0341                  | 0.0462                  | 0.0634                  | 0.0678                  |
| $R^2_{OOS}$  | -0.0302                 | -0.4034                 | -0.0154                 | -0.1758                 |

This table reports the regression results of the return on last half hour of the call options market,  $r_{LH}$ . The meaning of each dependent variable in the table is as follows:  $r_{ONFH}$  represents the return rate of buying from the closing price of the previous day and holding until thirty minutes before the opening of the day;  $r_{ROD}$  represents the return rate of buying from the closing price of the previous day and holding until thirty minutes before the closing of the day.  $r_M$  represents the return rate from buying and holding 30 minutes after the opening of the day to 60 minutes before closing;  $r_{SLH}$  represents the return rate from buying and holding 60 minutes before closing to 30 minutes before closing;  $R^2$  represents the R-squared statistic of linear regression; and  $R^2_{OOS}$  represents the R-squared statistic of the out-of-sample test. The sample interval is from January 2017 to January 2022. The significance levels of 1%,

5%, and 10% are marked by \*\*\*, \*\*, and \*, respectively.

In terms of the intraday effect of put options, the coefficients and significance of each momentum factor behave similarly to the call options results. The factors  $r_{ONFH}$ ,  $r_M$ , and  $r_{ROD}$  all show significant reversal effects within the sample, and the significance levels are consistent with those of call options. However, unlike for futures and call options, the predictive ability of  $r_{ONFH}$  is stronger than that of  $r_{ROD}$ : in model (4),  $r_{ONFH}$  has a higher coefficient and significance (-0.10 and 1% level significance), and the coefficient and significance of  $r_{ROD}$  are smaller (significant at -0.02 and 5% levels). In addition, in terms of out-of-sample performance, the intraday momentum factor of put options shows particularly significant out-of-sample predictive ability. The  $R_{OOS}^2$  of each factor is significantly greater than 0, and the combination of  $r_{ONFH}$  and  $r_{ROD}$  has the best out-of-sample prediction effect, with an  $R_{OOS}^2$  of 0.0321%, followed by the three-factor combination of  $r_{ONFH}$ ,  $r_M$ , and  $r_{ROD}$ . The previous performance is helpful in predicting the return rate of put options in the last half hour.

Table 3  
Commodity put options intraday momentum effect regression results

| PUT Option  | 1                       | 2                       | 3                       | 4                       |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|
| $r_{ONFH}$  | -0.1180***<br>(-5.0973) | -0.1230***<br>(-5.2994) |                         | -0.1006***<br>(-3.9815) |
| $r_M$       |                         | -0.0306**<br>(-2.5322)  |                         |                         |
| $r_{SLH}$   |                         | -0.0539<br>(-1.1940)    |                         |                         |
| $r_{ROD}$   |                         |                         | -0.0389***<br>(-3.6142) | -0.0202*<br>(-1.7167)   |
| $R^2$       | 0.1289                  | 0.1676                  | 0.0647                  | 0.1435                  |
| $R_{adj}^2$ | 0.1239                  | 0.1528                  | 0.0598                  | 0.1336                  |
| $R_{OOS}^2$ | 0.0283                  | 0.0290                  | 0.0240                  | 0.0321                  |

This table reports the regression results of the return on the last half hour of the put option market,  $r_{LH}$ . The meaning of each dependent variable in the table is as follows:  $r_{ONFH}$  represents the return rate of buying from the closing price of the previous day and holding until thirty minutes before the opening of the day;  $r_{ROD}$  represents the return rate of buying from the closing price of the previous day and holding until thirty minutes before the closing of the day.  $r_M$  represents the return rate from buying and holding 30 minutes after the opening of the day to 60 minutes before closing;  $r_{SLH}$  represents the return rate from buying and holding 60 minutes before closing to 30 minutes before closing;  $R^2$  represents the R-squared statistic of linear regression; and  $R_{OOS}^2$  represents the R-squared statistic of the out-of-sample test. The sample interval is from January 2017 to January 2022. The significance levels of 1%, 5%, and 10% are marked by \*\*\*, \*\*, and \*, respectively.

### 3.2 Liquidity and Intraday Reversal

In the Chinese market, both futures and options show significant intraday reversal effects. However, this type of reversal effect is not consistent with the findings of Gao et al. (2018). We believe that this is due to the market-maker system of China's futures and options markets. Various market makers exist in China's options and futures markets. Their duty is to provide liquidity to the market. The exchange regularly assesses the market makers' matching volume and provides certain rewards. Therefore, domestic market makers usually increase market liquidity by actively matching transactions during the day and accumulate directional positions in the process. While market makers provide market liquidity, they also need to control risk exposure. Therefore, when the market closes down, if the market liquidity is sufficient, market makers tend to close their positions to control their own risk indicators. If this assumption is true, the market will show a reversal effect under high liquidity and a trend effect under low liquidity.

Unfortunately, the current Chinese exchanges do not disclose the positions and transactions of market makers, and we cannot verify this phenomenon from specific positions. In response to the above conjecture, only market trading volume is available to conduct a difference analysis. From the perspective of a single product, we calculate the 10th percentile and 90th percentile of the daily trading volume of all orders placed in the sample and record the daily data of a product with a trading volume lower than the 10th percentile as a low trading volume group. The daily data of a product with a trading volume greater than or equal to the 90th percentile is defined as a high trading volume group. All the varieties are combined to generate a low trading volume group and a high trading volume group, and pooled regression analysis is performed.

Table 4. reports the regression results of the intraday momentum effect of the futures high-liquidity group and the low-liquidity group. The high-liquidity group shows a significant intraday reversal effect. The significance of each coefficient is similar to the results of the full sample, while the low-liquidity group shows a weak intraday momentum effect: only the  $r_{SLH}$  of the second group of regressions shows a significant positive correlation. In fact, in a high-liquidity environment, market makers have the conditions to close positions that are passively traded during the day, which will lead to a reversal effect, while in a low-liquidity environment, the positions that are passively traded throughout the day will be closed at the end of the day. In this scenario, liquidity is poor, and

market makers are unable to close positions. Instead, they continue to trade passively, resulting in a weak intraday momentum effect.

Table 4  
Liquidity and intraday momentum (futures)

|            | Panel A: High Turnover Future |           |           |           | Panel B: Low Turnover Future |          |         |         |
|------------|-------------------------------|-----------|-----------|-----------|------------------------------|----------|---------|---------|
| $r_{ONFH}$ | -0.025**                      | -0.028*** |           | -0.006    | 0.006                        | 0.008    |         | 0.004   |
|            | (-2.383)                      | (-2.680)  |           | (-0.512)  | (0.527)                      | (0.630)  |         | (0.282) |
| $r_M$      |                               | -0.022*** |           |           |                              | 0.000    |         |         |
|            |                               | (-3.578)  |           |           |                              | (0.049)  |         |         |
| $r_{SLH}$  |                               | -0.102*** |           |           |                              | 0.071*** |         |         |
|            |                               | (-4.397)  |           |           |                              | (3.417)  |         |         |
| $r_{ROD}$  |                               |           | -0.027*** | -0.026*** |                              |          | 0.004   | 0.003   |
|            |                               |           | (-4.908)  | (-4.318)  |                              |          | (0.737) | (0.587) |
| $R^2$      | 0.161                         | 1.090     | 0.681     | 0.689     | 0.009                        | 0.369    | 0.017   | 0.019   |

This table reports the regression results of intraday momentum effects for high-liquidity and low-liquidity groups of futures. The meaning of each dependent variable in the table is as follows:  $r_{ONFH}$  represents the return rate of buying from the closing price of the previous day and holding until thirty minutes before the opening of the day;  $r_{ROD}$  represents the return rate of buying from the closing price of the previous day and holding until thirty minutes before the closing of the day.  $r_M$  represents the return rate from buying and holding 30 minutes after the opening of the day to 60 minutes before closing;  $r_{SLH}$  represents the return rate from buying and holding 60 minutes before closing to 30 minutes before closing;  $R^2$  represents the R-squared statistic of linear regression; and  $R_{OOS}^2$  represents the R-squared statistic of the out-of-sample test. The sample interval is from January 2017 to January 2022. The significance levels of 1%, 5%, and 10% are marked by \*\*\*, \*\*, and \*, respectively.

Table 5. reports the intraday momentum effect regression results for the high-liquidity and low-liquidity groups of options. Considering that the price movements of call options and put options interfere with each other, we divided the data into call options groups and put options groups. The liquidity characteristics of options are similar to those of futures: the high-liquidity group shows a significant intraday reversal effect, and the significance of each coefficient is similar to the results of the full sample, while the low-liquidity group shows a relatively weak intraday reversal effect. This feature holds true for both call and put groups.

The futures and options results indicate that the high and low liquidity groups showed significant differences, and the direction and significance of the differences were in line with the hedging logic of market makers. It's noteworthy that despite the presence of market makers in the US market, futures exhibit an intraday momentum effect. This difference might be attributed to US

market makers engaging in the simultaneous market making of diverse, low-correlated assets, resulting in less stringent risk control per individual asset<sup>2</sup>. In contrast, the majority of market makers in the futures market specialize in handling futures contracts, which enforces stricter exposure requirements due to the singular asset focus. This constraint also drives Chinese market makers to close accumulated positions during periods of sufficient liquidity.

Table 5

Liquidity and intraday momentum (Options)

|            | Panel A: High Turnover Call |                       |                       |                       | Panel B: High Turnover Put |                       |                       |       |
|------------|-----------------------------|-----------------------|-----------------------|-----------------------|----------------------------|-----------------------|-----------------------|-------|
| $r_{ONFH}$ | -0.010<br>(-0.736)          | -0.013<br>(-1.015)    | 0.012<br>(0.830)      |                       | -0.055<br>(-2.997)         | -0.056***<br>(-3.016) | -0.037*<br>(-1.917)   |       |
| $r_M$      |                             | -0.027***<br>(-3.243) |                       |                       |                            | -0.020*<br>(-2.095)   |                       |       |
| $r_{SLH}$  |                             | -0.077***<br>(-2.698) |                       |                       |                            | -0.142***<br>(-3.482) |                       |       |
| $r_{ROD}$  |                             |                       | -0.025***<br>(-3.406) | -0.027***<br>(-3.428) |                            | -0.032<br>(-3.693)    | -0.026***<br>(-2.882) |       |
| $R^2$      | 0.023                       | 0.801                 | 0.499                 | 0.529                 | 0.743                      | 2.154                 | 1.125                 | 1.427 |
|            | Panel C: Low Turnover Call  |                       |                       |                       | Panel D: Low Turnover Put  |                       |                       |       |
| $r_{ONFH}$ | -0.029<br>(-1.383)          | -0.028<br>(-1.336)    | -0.034<br>(-1.546)    |                       | 0.034**<br>(2.408)         | 0.036**<br>(2.483)    | 0.036**<br>(2.251)    |       |
| $r_M$      |                             | -0.004<br>(-0.517)    |                       |                       |                            | 0.007<br>(0.826)      |                       |       |
| $r_{SLH}$  |                             | 0.148***<br>(4.006)   |                       |                       |                            | 0.016<br>(0.649)      |                       |       |
| $r_{ROD}$  |                             |                       | 0.002<br>(0.272)      | 0.005<br>(0.743)      |                            | 0.007<br>(0.877)      | -0.002<br>(-0.210)    |       |
| $R^2$      | 0.156                       | 1.458                 | 0.006                 | 0.201                 | 0.288                      | 0.341                 | 0.038                 | 0.290 |

This table reports the regression results of intraday momentum effects for high-liquidity and low-liquidity groups of options. The meaning of each dependent variable in the table is as follows:  $r_{ONFH}$  represents the return rate of buying from the closing price of the previous day and holding until thirty minutes before the opening of the day;  $r_{ROD}$  represents the return rate of buying from the closing price of the previous day and holding until thirty minutes before the closing of the day.  $r_M$  represents the return rate from buying and holding 30 minutes after the opening of the day to 60 minutes before closing;  $r_{SLH}$  represents the return rate from buying and holding 60 minutes before closing to 30 minutes before closing;  $R^2$  represents the R-squared statistic of linear regression; and  $R_{DOS}^2$  represents the R-squared statistic of the out-of-sample test. The sample interval is from January 2017 to January 2022. The significance levels of 1%, 5%, and 10% are marked by \*\*\*, \*\*, and \*, respectively.

<sup>2</sup> <https://m.gelonghui.com/p/544732> : 'The Development Path and Experience of the U.S. Market Making Business'

### 3.3 Cross-Sectional Effect between futures and options

As mentioned previously, commodity futures are the underlying asset of commodity options, and both show significant intraday reversal effects. This section will examine whether a cross-prediction effect exists between options and futures.

#### 3.3.1 Cross-forecasting effect of options on futures

This section examines the cross-prediction effect of options on futures. Table 6. and Table 7. report the regression results of commodity call options/put options on commodity futures returns in the last half hour, respectively. The superscript of  $r_*^O$  in the table represents the asset class to which it belongs: O represents options, and F represents commodities. For each table, we conducted 8 sets of regressions. The penultimate row in the table indicates the factors tested by the model; the last row in the table indicates whether the regression model controls its own factors. For example, in Table 3.6, models (1) and (2) test the ability of  $r_{ONFH}^O$  of a call option to predict the last half-hour return of the futures, among which model (1) includes only the option factor, that is,  $r_{ONFH}^O$ ; model (2) controls both factors of futures and options, namely,  $r_{ONFH}^O$  and  $r_{ONFH}^F$ .

Table 6. reports the results of the regression of the call options momentum factor on the last half-hour futures returns. Model (1) and model (2) show that  $r_{ONFH}^O$  has a significant predictive ability for the last half-hour return of futures, and after adding  $r_{ONFH}^F$  as a control variable, it is still significant at the 5% significance level. In addition, at the single-factor level,  $r_{ONFH}^O$  has excellent out-of-sample prediction performance for the last half-hour return of futures, but after controlling for  $r_{ONFH}^F$ , the out-of-sample prediction ability decreases. Model (3) and model (4) test the predictive ability of  $r_{ONFH}^O$ ,  $r_M^O$  and  $r_{SLH}^O$  on futures returns in the last half hour. Without controlling the futures' own factors, the three factors are all significant at the 1% level and have excellent out-of-sample prediction ability; after controlling the futures' own factors,  $r_{ONFH}^O$  and  $r_{SLH}^O$  remain significant at the 5% level, while  $r_M^O$  is no longer significant, and the out-of-sample prediction effect is poor. The results of model (5) and model (6) show that  $r_{ROD}^F$  has significant in-sample (both significant at the 1% level) and out-of-sample predictive ability ( $R_{OOS}^2$  is greater than 0). Model (7) and model (8) compared  $r_{ONFH}^O$  and  $r_{ROD}^O$  and found that  $r_{ROD}^O$  has stronger predictive ability. Overall, call options have cross-predictive ability on futures returns, and this effect is manifested as a reversal effect. From the perspective of subdivision factors,  $r_{ROD}^O$  has the best out-of-sample prediction performance.

Table 7. reports the results of the regression of the put option's momentum factor on the last



half hour of futures returns, and the results are similar to those for call options. In terms of significance and out-of-sample predictive ability,  $r_{ONFH}^O$ ,  $r_M^O$  and  $r_{SLH}^O$  of the put option show higher significance and better out-of-sample predictive ability without controlling the futures' own momentum factor, but after adding the futures momentum factor, the significance disappears, and the out-of-sample predictive ability decreases.  $r_{ROD}^O$  shows continued significance and out-of-sample predictive ability. However, differences remain between the two, mainly reflected in the differences in coefficients. The coefficient of call options on futures returns in the last half hour is negative, showing a reversal effect, while the regression coefficient of put options is overall positive, indicating that when put options obtain positive returns, futures returns in the last half hour will also be positive. However, the source of positive returns from put options is more complex. They may be due to price increases caused by an increase in intrinsic value (futures fall) or vega returns brought about by increased volatility. Judging from the results of the control group (controlling the momentum factor of the futures itself offsets the change in the intrinsic value of the option to some extent), the ability of put options to predict futures' returns in the last half hour comes from the change in the intrinsic value of the option.

Table 6

## Cross Momentum Effect: Regression Results of Call Options on Futures

| $r_{LH, CALL}^F$ | 1                       | 2                      | 3                       | 4                      | 5                       | 6                       | 7                       | 8                      |
|------------------|-------------------------|------------------------|-------------------------|------------------------|-------------------------|-------------------------|-------------------------|------------------------|
| $r_{ONFH}^O$     | -0.0006***<br>(-4.2052) | -0.0004**<br>(-2.1677) | -0.0008***<br>(-5.0198) | -0.0005**<br>(-2.5399) |                         |                         | -0.0004**<br>(-2.1150)  | -0.0002<br>(-0.8954)   |
| $r_M^O$          |                         |                        | -0.0004***<br>(-4.0301) | -0.0003*<br>(-1.8268)  |                         |                         |                         |                        |
| $r_{SLH}^O$      |                         |                        | -0.0012***<br>(-3.7339) | -0.0011**<br>(-2.4946) |                         |                         |                         |                        |
| $r_{ROD}^O$      |                         |                        |                         |                        | -0.0005***<br>(-6.3523) | -0.0004***<br>(-3.0453) | -0.0004***<br>(-4.9073) | -0.0003**<br>(-2.3290) |
| $r_{ONFH}^F$     |                         | -0.0107**<br>(-2.1021) |                         | -0.0121**<br>(-2.2494) |                         |                         |                         | -0.0083<br>(-1.4094)   |
| $r_M^F$          |                         |                        |                         | -0.0036<br>(-1.0325)   |                         |                         |                         |                        |
| $r_{SLH}^F$      |                         |                        |                         | -0.0035<br>(-0.3510)   |                         |                         |                         |                        |
| $r_{ROD}^F$      |                         |                        |                         |                        |                         | -0.0052*<br>(-1.7465)   |                         | -0.0038<br>(-1.1669)   |
| $R^2$            | 0.0859                  | 0.1107                 | 0.2551                  | 0.2833                 | 0.2013                  | 0.2154                  | 0.2241                  | 0.2536                 |
| $R_{adj}^2$      | 0.0811                  | 0.1009                 | 0.2401                  | 0.2532                 | 0.1963                  | 0.2054                  | 0.2141                  | 0.2335                 |
| $R_{OOS}^2$      | 0.0499                  | -0.1324                | 0.1372                  | -0.2742                | 0.1828                  | 0.0751                  | 0.1639                  | -0.1351                |

This table reports the regression results of commodity call options on commodity futures  $r_{LH}^F$ . The meaning of the subscripts is consistent with the previous chapter; the superscript O represents options, and F represents futures.  $R^2$  represents the R-squared statistic of linear regression;  $R_{OOS}^2$  represents the R-squared statistic of out-of-sample testing. The sample interval is from January 2017 to January 2022. The significance levels of 1%, 5%, and 10% are marked by \*\*\*, \*\*, and \*, respectively. The Test Variable in the penultimate line indicates the purpose of the regression, and the Self Control in the last line indicates whether the regression controls its own momentum factor.

Table 7

## Cross Momentum Effect: Regression Results of Put Options on Futures

| $r_{LH, PUT}^F$ | 1                     | 2                       | 3                     | 4                       | 5                     | 6                       | 7                     | 8                       |
|-----------------|-----------------------|-------------------------|-----------------------|-------------------------|-----------------------|-------------------------|-----------------------|-------------------------|
| $r_{ONFH}^O$    | 0.0005***<br>(3.3076) | 0.0002<br>(0.9049)      | 0.0007***<br>(4.3114) | 0.0003<br>(1.4772)      |                       |                         | 0.0004**<br>(2.3833)  | 0.0003<br>(1.2040)      |
| $r_M^O$         |                       |                         | 0.0003***<br>(2.8264) | 0.0000<br>(0.8855)      |                       |                         |                       |                         |
| $r_{SLH}^O$     |                       |                         | 0.0009***<br>(2.6995) | 0.0004<br>(0.9711)      |                       |                         |                       |                         |
| $r_{ROD}^O$     |                       |                         |                       |                         | 0.0004***<br>(4.6871) | 0.0001<br>(0.3424)      | 0.0003***<br>(3.3243) | 0.0000<br>(0.7376)      |
| $r_{ONFH}^F$    |                       | -0.0147***<br>(-2.8717) |                       | -0.0161***<br>(-2.9343) |                       |                         |                       | -0.0068<br>(-1.1480)    |
| $r_M^F$         |                       |                         |                       | -0.0085***<br>(-2.7065) |                       |                         |                       |                         |
| $r_{SLH}^F$     |                       |                         |                       | -0.0163*<br>(-1.6663)   |                       |                         |                       |                         |
| $r_{ROD}^F$     |                       |                         |                       |                         |                       | -0.0099***<br>(-3.5610) |                       | -0.0090***<br>(-2.9587) |
| $R^2$           | 0.0532                | 0.0943                  | 0.1586                | 0.2415                  | 0.1098                | 0.1737                  | 0.1376                | 0.2148                  |
| $R_{adj}^2$     | 0.0483                | 0.0845                  | 0.1436                | 0.2114                  | 0.1048                | 0.1636                  | 0.1276                | 0.1947                  |
| $R_{OOS}^2$     | 0.0221                | -0.0815                 | -0.0009               | -0.2465                 | 0.0443                | 0.0646                  | 0.0386                | -0.0976                 |

This table reports the regression results of commodity put options on commodity futures  $r_{LH}^F$ . The meaning of the subscripts is consistent with the previous chapter; the superscript O represents options, and F represents futures.  $R^2$  represents the R-squared statistic of linear regression;  $R_{OOS}^2$  represents the R-squared statistic of out-of-sample testing. The sample interval is from January 2017 to January 2022. The significance levels of 1%, 5%, and 10% are marked by \*\*\*, \*\*, and \*, respectively. The Test Variable in the penultimate line indicates the purpose of the regression, and the Self Control in the last line indicates whether the regression controls its own momentum factor.

Table 8

## Cross Momentum Effect: Regression Results of Futures on Call Options

| $r_{LH, CALL}^O$ | 1                       | 2                    | 3                       | 4                       | 5                     | 6                       | 7                      | 8                       |
|------------------|-------------------------|----------------------|-------------------------|-------------------------|-----------------------|-------------------------|------------------------|-------------------------|
| $r_{ONFH}^F$     | -0.4960***<br>(-2.9791) | -0.3478<br>(-1.5975) | -0.5087***<br>(-2.9924) | -0.3023<br>(-1.3645)    |                       |                         | -0.4429**<br>(-2.4574) | -0.5418**<br>(-2.2397)  |
| $r_M^F$          |                         |                      | -0.0033<br>(-0.0361)    | 0.2952**<br>(2.0650)    |                       |                         |                        |                         |
| $r_{SLH}^F$      |                         |                      | -0.9141***<br>-2.9999   | -1.9736***<br>(-4.7896) |                       |                         |                        |                         |
| $r_{ROD}^F$      |                         |                      |                         |                         | -0.1541*<br>(-1.8747) | 0.1668<br>(1.3735)      | -0.0769<br>(-0.8735)   | 0.2743**<br>(2.0266)    |
| $r_{ONFH}^O$     |                         | -0.0101<br>(-1.1936) |                         | -0.0121<br>(-1.4036)    |                       |                         |                        | 0.0043<br>(0.4439)      |
| $r_M^O$          |                         |                      |                         | -0.0171***<br>(-2.8419) |                       |                         |                        |                         |
| $r_{SLH}^O$      |                         |                      |                         | 0.0684***<br>(3.9443)   |                       |                         |                        |                         |
| $r_{ROD}^O$      |                         |                      |                         |                         |                       | -0.0179***<br>(-3.6035) |                        | -0.0193***<br>(-3.3993) |
| $R^2$            | 0.0435                  | 0.0527               | 0.0880                  | 0.2216                  | 0.0174                | 0.0822                  | 0.0473                 | 0.1164                  |
| $R_{adj}^2$      | 0.0386                  | 0.0427               | 0.0732                  | 0.1915                  | 0.0124                | 0.0721                  | 0.0374                 | 0.0963                  |
| $R_{OOS}^2$      | -0.0153                 | -0.0966              | -0.0600                 | -0.5343                 | -0.0159               | -0.1530                 | -0.0754                | -0.4020                 |

This table reports the regression results of commodity futures on call options  $r_{LH}^O$ . The meaning of the subscripts is consistent with the previous chapter; the superscript O represents options, and F represents futures.  $R^2$  represents the R-squared statistic of linear regression;  $R_{OOS}^2$  represents the R-squared statistic of out-of-sample testing. The sample interval is from January 2017 to January 2022. The significance levels of 1%, 5%, and 10% are marked by \*\*\*, \*\*, and \*, respectively. The Test Variable in the penultimate line indicates the purpose of the regression, and the Self Control in the last line indicates whether the regression controls its own momentum factor.

Table 9

## Cross Momentum Effect: Regression Results of Futures on PUT Options

| $r_{LH, PUT}^O$ | 1                  | 2                       | 3                    | 4                       | 5                   | 6                       | 7                  | 8                       |
|-----------------|--------------------|-------------------------|----------------------|-------------------------|---------------------|-------------------------|--------------------|-------------------------|
| $r_{ONFH}^F$    | 0.8861<br>(1.6155) | -1.5546**<br>(-2.1251)  | 0.9637*<br>(1.7199)  | -1.6899**<br>(-2.2698)  |                     |                         | 0.6596<br>(1.1104) | -1.4914**<br>(-1.8552)  |
| $r_M^F$         |                    |                         | 0.1478<br>(0.4908)   | -0.7599*<br>(-1.7813)   |                     |                         |                    |                         |
| $r_{SLH}^F$     |                    |                         | 2.4714**<br>(2.4602) | 2.6953**<br>(2.0263)    |                     |                         |                    |                         |
| $r_{ROD}^F$     |                    |                         |                      |                         | 0.4462*<br>(1.6485) | -0.4350<br>(-1.1532)    |                    | -0.0441<br>(-0.1068)    |
| $r_{ONFH}^O$    |                    | -0.1604***<br>(-5.2625) |                      | -0.1680***<br>(-5.4622) |                     |                         |                    | -0.1422***<br>(-4.1669) |
| $r_M^O$         |                    |                         |                      | -0.0516***<br>(-3.0271) |                     |                         |                    |                         |
| $r_{SLH}^O$     |                    |                         |                      | 0.0211<br>(0.3556)      |                     |                         |                    |                         |
| $r_{ROD}^O$     |                    |                         |                      |                         |                     | -0.0511***<br>(-3.4135) | 0.3312<br>(1.1428) | -0.0209<br>(-1.2523)    |
| $R^2$           | 0.0128             | 0.1523                  | 0.0441               | 0.2303                  | 0.0135              | 0.0722                  | 0.0196             | 0.1670                  |
| $R_{adj}^2$     | 0.0079             | 0.1423                  | 0.0292               | 0.2001                  | 0.0085              | 0.0621                  | 0.0097             | 0.1469                  |
| $R_{OOS}^2$     | 0.0021             | 0.0238                  | 0.0386               | 0.0658                  | 0.0142              | 0.0162                  | 0.0125             | 0.0250                  |

This table reports the regression results of commodity futures on call options  $r_{LH}^O$ . The meaning of the subscripts is consistent with the previous chapter; the superscript O represents options, and F represents futures.  $R^2$  represents the R-squared statistic of linear regression;  $R_{OOS}^2$  represents the R-squared statistic of out-of-sample testing. The sample interval is from January 2017 to January 2022. The significance levels of 1%, 5%, and 10% are marked by \*\*\*, \*\*, and \*, respectively. The Test Variable in the penultimate line indicates the purpose of the regression, and the Self Control in the last line indicates whether the regression controls its own momentum factor.

### 3.3.2 Cross-forecasting effect of futures on options

This section examines the cross-prediction effect of futures on options. Table 8. and Table 9. report the regression results of commodity futures on the last half-hour returns of commodity call/put options, respectively. The superscript of  $r_*^O$  in the table represents the asset class to which it belongs: O represents options, and F represents commodities. For each table, we conducted 8 sets of regressions, and the bottom two rows in the table have the same meaning as before.

Table 8. reports the results of the regression of the futures momentum factor on the last half-hour returns of call options. In model (1), the futures factor  $r_{ONFH}^F$  is significant at the 1% level, but the significance disappears after controlling for its own variables. Similarly,  $r_{ROD}^F$  is significant in the single-factor regression, but the significance disappears after controlling the variables (models (5, 6)). However, in models (3) and (4),  $r_{SLH}^F$  is significant, and the significance is maintained at the 1% level after controlling for  $r_{SLH}^O$ . In the comparison of  $r_{ONFH}^F$  and  $r_{ROD}^F$ ,  $r_{ONFH}^F$  showed strong significance. However, the out-of-sample forecasting ability is weak, and no group is better than the mean forecast, indicating that the futures momentum factor has weak ability to predict futures returns in the last half hour.

The results in Table 9. are quite different from those in Table 8.: the futures momentum factor has significant predictive power for the last half-hour return of put options: Models (1) – Model (8) all report positive  $R_{OOS}^2$ . In addition, the significance level of the  $r_{ONFH}^F$  single factor is low, but after adding its own factor, the significance level increases, the coefficient changes sign, and the model  $R^2$  increases. A similar situation occurs in the three-factor model of  $r_{ONFH}^F$ ,  $r_M^F$  and  $r_{SLH}^F$ , in which the coefficient of  $r_{ONFH}^F$  changes from 0.96 to -1.69, and the t value changes from 1.72 to -2.27; the coefficient of  $r_M^F$  changes from 0.15 to -0.76, and the t value changes from 0.49 to -1.78; and the coefficient and significance of  $r_{SLH}^F$  do not change substantially. The change in the sign of the factor coefficient can be considered from the perspective of the value composition of the option. The value of an option can be regarded as the sum of the time value and intrinsic value. Without considering the option at the end of the day, the time value of the option is related to the volatility of the futures and the duration of the option. When the volatility is relatively stable, the time value of the option may not experience substantial changes within days. Therefore, after incorporating the option's own momentum factor, the change in the coefficient shows that the momentum of the futures has a high-order impact on the last half-hour return of the put option. The results of Model (5)-Model (8) show

that the predictive ability of  $r_{ROD}^F$  is weaker than that of  $r_{ONFH}^F$ . The results in Table 9. show that futures prices have significant predictive ability for the last half-hour return of put options, and this predictive ability does not disappear after controlling for its own variables. In summary, the cross-predictive ability of futures to options is reflected mainly on the put options side, while the predictive ability on the call options side is weak.

### 3.4 Vega Hedge: Explanation of cross effects between futures and options

The underlying commodity options in China's options market are generally commodity futures, so changes in futures also drive changes in options. Based on the perspective of options hedging, if options traders need to hedge delta and gamma risks, they will purchase a proportion of futures positions to offset the delta and gamma exposures of options. If the volatility of futures changes drastically, the Vega exposure of options traders will increase. Trading in the options market is an effective way to manage Vega exposure.

When options traders observe amplified volatility in futures, they perform corresponding operations on options to manage Vega risk. This section verifies this conjecture by imitating the calculation method of the momentum factor and calculating the corresponding volatility momentum factor:

$$Vol_{ONFH,t} = \frac{V_{o:o+30,t}}{V_{c-30:c,t-1}} - 1$$

$V_{o:o+30,t}$  represents the 1-minute annualized volatility from the opening of the day to 30 minutes before the opening of the day, and  $V_{c-30:c,t-1}$  represents the 1-minute annualized volatility from the opening of the day to 30 minutes before the closing of the previous trading day.

$$Vol_{ROD,t} = \frac{V_{30:c-30,t}}{V_{c-30:c,t-1}} - 1$$

$V_{30:c,t}$  represents the 1-minute annualized volatility from 30 minutes after the opening of the day to 30 minutes before the close of the day, and  $V_{c-30:c,t-1}$  represents the 1-minute annualized volatility from 30 minutes before the close of the previous trading day to close.

$$Vol_{M,t} = \frac{V_{30:c-60,t}}{V_{o:o+30,t}} - 1$$

$V_{c-30:c-60,t}$  represents the 1-minute annualized volatility from 30 minutes before the opening of the day to 60 minutes before the closing of the day, and  $V_{o:o+30,t}$  represents the 1-minute annualized volatility from 30 minutes before the closing of the day to the close.

$$Vol_{SLH,t} = \frac{V_{c-90:c-60,t}}{V_{c-30:c,t}} - 1$$

$V_{c-90:c-60,t}$  represents the 1-minute annualized volatility from 90 minutes before the close of the day to 60 minutes before the close of the day, and  $V_{c-30:c,t}$  represents the 1-minute annualized volatility from 30 minutes before the close of the day to close.

Table 10  
Futures Volatility Momentum and Options Return

|                 | 1                    | 2                    | 3                  |
|-----------------|----------------------|----------------------|--------------------|
| $F_{vol\_ONFH}$ | 0.0062**<br>(2.5427) | 0.0054**<br>(1.9596) |                    |
| $F_{vol\_M}$    |                      | -0.0022<br>(-0.7973) |                    |
| $F_{vol\_SLH}$  |                      | 0.0078<br>(1.5752)   |                    |
| $F_{vol\_ROD}$  |                      |                      | 0.0004<br>(0.1274) |
| $R^2$           | 0.0167               | 0.0244               | 0.0000             |
| $R^2_{OOS}$     | 0.0792               | 0.0687               | -0.1036            |

This table reports the intraday momentum effect regression results of options return momentum and futures volatility momentum. The explained variable is  $O_{price\_LH}$ , which represents the last half-hour return rate of the option. The meaning of each dependent variable is as follows:  $F_{vol\_ONFH}$  represents, for the current day, the momentum of the volatility in the 30 minutes before the opening and the volatility in the 30 minutes before the closing of yesterday;  $F_{vol\_ROD}$  represents the momentum of the volatility in the 30 minutes before the closing of the day and the volatility of the 30 minutes before yesterday's closing;  $F_{vol\_M}$  represents the momentum of the volatility 30 minutes after the opening of the day, the momentum of the volatility one hour before the close and the volatility 30 minutes before the opening of the day;  $F_{vol\_SLH}$  represents the momentum of the volatility from 60 minutes before the close to 30 minutes before the close and the volatility 30 minutes before the close of the day.

Table 10. reports the momentum regression results of futures volatility and options returns. The momentum effect between Chinese futures and options is highly significant.  $F_{vol\_ONFH}$  has a positive predictive effect on the half-hour returns of options before closing in Model (1) and Model (2). Its coefficients are 0.0062 and 0.0054, respectively, and both are significant at the 10% level. Moreover, Model (1) and the out-of-sample  $R^2_{OOS}$  are all greater than 0 (0.0792, 0.0687), indicating that the prediction effect of the model is better than the mean prediction. Other variables have no predictive effect, and their coefficients are not significant. The results in Table 10. show that Vega hedge behavior exists in the Chinese commodity market. The intraday volatility changes in futures can effectively predict the returns of options in the last half hour.



#### 4. Conclusion

Based on 1-minute high-frequency data from China's commodity futures and options market from 2017 to 2022, we examine the intraday momentum effect of China's commodity futures and options. China's options and futures markets have significant intraday reversal effects, and the overnight and opening factors (ONFH) and intraday factors (ROD) are predictive of the market's return in the last half hour (LH). By comparing the ONFH and the ROD, we find that in most cases (futures, call options), the ROD factor is a better predictor, but for put options, the predictive ability of the overnight opening factor is more significant. More importantly, we identify a novel economic force that drives market intraday reversal: market makers' exposure management constraints prompt them to close positions near the daily close. Our tests reveal that ample market liquidity corresponds to a notable intraday reversal effect in China's commodity futures and options. Conversely, when market liquidity is scarce, these markets display a significant intraday trend effect, albeit with reduced significance. This pattern aligns with the discernible trading behavior of market makers.

Additionally, we tested the intraday predictive capability between futures and options. We find a strong cross-predictive ability from futures to options, while the cross-predictive ability of options for futures is weak. Regarding the source of the cross effect, we start from the trading behavior of option traders and find that the cross effect is caused by the vega hedging behavior of options traders. We construct a series of futures volatility momentum factors and test their ability to predict the last half-hour returns of options. Our tests show that the volatility ONFH factor can effectively predict the return of options in the last half hour.

There are some unanswered questions related to intraday momentum. First, the empirical facts reported in this article require new theoretical models of day trading to identify the factors that determine its risk premium. Second, we focus on China's commodity futures and options market but does not examine China's stock index futures and options, which are also worthy of study. Third, there is a large literature on predictability at monthly frequencies, but the relationship between intraday predictability and monthly predictability is unclear. These would be fruitful topics for future research.

## Appendix A. Details on Futures Contracts

Table A1

| Future Code | Commodity               | Sector                | Has Options |
|-------------|-------------------------|-----------------------|-------------|
| AL          | Aluminum                | Non-Ferrous Metals    | YES         |
| CU          | Copper                  | Non-Ferrous Metals    | YES         |
| NI          | Nickel                  | Non-Ferrous Metals    | NO          |
| PB          | Lead                    | Non-Ferrous Metals    | NO          |
| SN          | Tin                     | Non-Ferrous Metals    | NO          |
| ZN          | Zinc                    | Non-Ferrous Metals    | YES         |
| A           | Soybean No. 1           | Agricultural Products | YES         |
| AP          | Apple                   | Agricultural Products | NO          |
| B           | Soybean No. 2           | Agricultural Products | YES         |
| C           | Corn                    | Agricultural Products | YES         |
| CF          | Cotton                  | Agricultural Products | YES         |
| CJ          | Red Dates               | Agricultural Products | NO          |
| CS          | Corn Starch             | Agricultural Products | NO          |
| CY          | Cotton Yarn             | Agricultural Products | NO          |
| FB          | Fiberboard              | Agricultural Products | NO          |
| JD          | Egg                     | Agricultural Products | NO          |
| M           | Soybean Meal            | Agricultural Products | YES         |
| OI          | Canola Oil              | Agricultural Products | YES         |
| P           | Palm Oil                | Agricultural Products | YES         |
| PK          | Peanut                  | Agricultural Products | YES         |
| RM          | Rapeseed Meal           | Agricultural Products | YES         |
| RR          | Japonica Rice           | Agricultural Products | NO          |
| RS          | Rapeseed                | Agricultural Products | NO          |
| SR          | White Sugar             | Agricultural Products | YES         |
| Y           | Soybean Oil             | Agricultural Products | YES         |
| BU          | Asphalt                 | Energy and Chemical   | NO          |
| EB          | Styrene                 | Energy and Chemical   | NO          |
| EG          | Ethylene Glycol         | Energy and Chemical   | NO          |
| FG          | Glass                   | Energy and Chemical   | NO          |
| FU          | Fuel Oil                | Energy and Chemical   | NO          |
| L           | Polyethylene            | Energy and Chemical   | YES         |
| LU          | Low Sulfur Fuel Oil     | Energy and Chemical   | NO          |
| MA          | Methanol                | Energy and Chemical   | YES         |
| NR          | Rubber 20               | Energy and Chemical   | NO          |
| PF          | Staple Fiber            | Energy and Chemical   | NO          |
| PG          | Liquefied Petroleum Gas | Energy and Chemical   | YES         |
| PP          | Polypropylene           | Energy and Chemical   | YES         |
| RU          | Natural Rubber          | Energy and Chemical   | YES         |
| SA          | Soda Ash                | Energy and Chemical   | NO          |
| SC          | Crude                   | Energy and Chemical   | YES         |
| SP          | Pulp                    | Energy and Chemical   | NO          |
| TA          | Pta                     | Energy and Chemical   | YES         |
| UR          | Urea                    | Energy and Chemical   | NO          |
| V           | Pvc                     | Energy and Chemical   | YES         |
| HC          | Hot Roll                | Black Metal           | NO          |
| I           | Iron Ore                | Black Metal           | YES         |

|    |                   |                |     |
|----|-------------------|----------------|-----|
| J  | Coke              | Black Metal    | NO  |
| JM | Coking Coal       | Black Metal    | NO  |
| RB | Rebar             | Black Metal    | YES |
| SF | Ferrosilicon      | Black Metal    | NO  |
| SM | Manganese Silicon | Black Metal    | NO  |
| SS | Stainless Steel   | Black Metal    | NO  |
| WR | Wire              | Black Metal    | NO  |
| AG | Silver            | Precious Metal | YES |
| AU | Gold              | Precious Metal | YES |

This Table shows the futures included in our sample. The first row is the code of the future in Chinese futures markets, the second row is the commodity name, and the third row shows the sector to which the future belongs. Most of the futures can be classified into 5 sectors: nonferrous metals, agricultural products, energy and chemicals, black metals, and precious metals. The last row indicates whether the future has options. These futures and options are used to elucidate the cross-sectional effects between the futures market and options market in China.

## References

- Asness, C. S., Moskowitz, T. J. and Pedersen, L. H. (2013) 'Value and Momentum Everywhere', *The Journal of Finance* 68(3): 929-85.
- Baltussen, G., Da, Z., Lammers, S. and Martens, M. (2021) 'Hedging Demand and Market Intraday momentum', *Journal of Financial Economics* 142(1): 377-403.
- Bogousslavsky, V. (2016) 'Infrequent Rebalancing, Return Autocorrelation, and Seasonality', *The Journal of Finance* 71(6): 2967-3006.
- Chang, Y. and Xue, J. (2018) 'The Empirical Research of Momentum and Reversal Effect on Stock Market of China——Analysis Based on Time Frequency and Market States', *Review of Investment Studies*.
- Chui, A., Ranganathan, K., Rohit, A. and Veeraraghavan, M. (2023) 'Momentum, Reversals and Liquidity: Indian Evidence', *Pacific-Basin Finance Journal* 82: 102193.
- Drew, M. E., Veeraraghavan, M., Ye, M. and Galagedera, D. U. A. (2007) 'Do Momentum Strategies Work? Australian Evidence', *Managerial Finance* 33(10): 772-87.
- Fan, J. H. and Qiao, X. (2023) 'Commodity Momentum: A Tale of Countries and Sectors', *Journal of Commodity Markets* 29: 100315
- Gao, L., Han, Y., Zhengzi Li, S. and Zhou, G. (2018) 'Market Intraday Momentum', *Journal of Financial Economics* 129(2): 394-414.
- Gao, Y., Han, X., Li, Y. and Xiong, X. (2019) 'Overnight Momentum, Informational Shocks, and Late Informed Trading in China', *International Review of Financial Analysis* 66: 101394.
- Gong, Y., Wang, M. and Dlugosch, D. (2019) 'Impacts of Ambiguity Aversion and Information Uncertainty on Momentum: An International Study', *Pacific-Basin Finance Journal* 55: 1-28.
- Heston, S., Korajczyk, R. and Sadka, R. (2010) 'Intraday Patterns in the Cross-section of Stock Returns', *The Journal of Finance* 65(4), 1369–1407.
- Ho, H., Hsiao, Y., Lo, W. and Yang, N. (2023) 'Momentum Investing and A tale of Intraday and Overnight Returns: Evidence from Taiwan', *Pacific-Basin Finance Journal* 82: 102151.
- Huang, W., Zhou, F., Yu, C., Hu, Y., Zhang, H. and Xu, Y. (2023) 'Momentum Effect and Contrarian Effect in China's A-Share Market, under Registration-Based System', *Pacific-Basin Finance Journal* 81: 102126.
- Jegadeesh, N. and Titman, S. (1993) 'Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency', *The Journal of Finance* 48(1): 65-91.
- Jegadeesh, N. and Titman, S. (2023) 'Momentum: Evidence and Insights 30 Years Later', *Pacific-Basin Finance Journal*: 102202.
- Jin, M., Kearney, F., Li, Y. and Yang, Y. C. (2020) 'Intraday Time-Series Momentum: Evidence from China', *The Journal of Futures Markets* 40(4): 632-50.
- Kang, J., Liu, M. H. and Ni, S. X. (2002) 'Contrarian and Momentum Strategies in the China Stock Market: 1993-2000', *Pacific-Basin Finance Journal* 10(3): 243-65.
- Kwon, K. Y., Kang, J. and Yun, J. (2020) 'Weekly Momentum in the Commodity Futures Market', *Finance Research Letters* 35: 101306.
- Li, C., Qiao, K. and Gao, B. (2015) 'Nonlinear Momentum Trading Strategies: Theory and Applications to China Futures Market', *Quarterly Journal of Finance*.
- Li, Y., Liang, C. and L. D. Huynh, T. (2022) 'A New Momentum Measurement in the Chinese Stock Market', *Pacific-Basin Finance Journal* 73: 101759.

- Li, Y., Shen, D., Wang, P. and Zhang, W. (2020) 'Does Intraday Time-Series Momentum Exist in Chinese Stock Index Futures Market?', *Finance Research Letters* 35: 101292.
- Ling A-Fan, Wang J-M, Yan W. (2017) 'Does A Good Opening Lead to A Good Closing--Empirical Evidence from CSI 300 Stock Index Futures', *Contemporary Finance & Economics*, (12):45-57.
- Limkriangkrai, M., Chai, D. and Zheng, G. (2023) 'Market Intraday Momentum: APAC Evidence', *Pacific-Basin Finance Journal* 80: 102086.
- Ming, L., Song, W. and Dong, M. (2023) 'Revisiting Time Series Momentum in China's Commodity Futures Market: Evidence on Sources of Momentum Profits', *Economic Modelling* 128: 106522.
- Naughton, T., Truong, C. and Veeraraghavan, M. (2008) 'Momentum Strategies and Stock Returns: Chinese Evidence', *Pacific-Basin Finance Journal* 16(4): 476-92.
- Rouwenhorst, K. G. (1998) 'International Momentum Strategies', *The Journal of Finance* 53(1): 267-84.
- Subrahmanyam, A. (2018) 'Equity Market Momentum: A Synthesis of the Literature and Suggestions for Future Work', *Pacific-Basin Finance Journal* 51: 291-6.
- Wang, R. and Ma, F. (2021) 'Intraday Return Predictability: Based on Intraday Jumps and Momentum', *Systems Engineering - Theory & Practice* 41(8): 2004-14.
- Wen, D., Wang, Y. and Zhang, Y. (2021) 'Intraday Return Predictability in China's Crude Oil Futures Market: New Evidence from A Unique Trading Mechanism', *Economic Modelling* 96: 209-19.
- Yan C. and Jiang X. (2018) 'An Empirical Study of Momentum and Reversal Effect in Chinese Stock Market--An Analysis Based on Time Frequency and Market State', *Investment Research* 37(02):74-86.
- Yang, N., Chu, H., Ko, K. and Lee, S. (2018) 'Continuing Overreaction and Momentum in A Market with Price Limits', *Pacific-Basin Finance Journal* 48: 56-71.
- Yin, L. and Wei, Y. (2020) 'Aggregate Profit Instability and Time Variations in Momentum Returns: Evidence from China', *Pacific-Basin Finance Journal* 60: 101276.
- Zhang, W., Wang, P. and Li, Y. (2020) 'Intraday Momentum in Chinese Commodity Futures Markets', *Research in International Business and Finance* 54: 101278.
- Zhang, X., Xiao, J. and Zhang, Z. (2020) 'An Anatomy of Commodity Futures Returns in China', *Pacific-Basin Finance Journal* 62: 101366.
- Zhang, Y., Ma, F. and Zhu, B. (2019) 'Intraday Momentum and Stock Return Predictability: Evidence from China', *Economic Modelling* 76: 319-29.
- Zhu, S., Zhou, C., Liu, H. and Ren, Y. (2023) 'Commodity Factor Investing via Machine Learning', *Pacific-Basin Finance Journal*: 102231.